NORMALITY IS THE CONSTANCY OF EMBEDDING IMAGES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: This characterization is the workhorse behind the lift-maps lemma (delivered L48358) and the automorphism-counting lemma (delivered L48349).

Editorial note. The label lemma-characterize-normal names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Chapter definition of normal: an algebraic extension E over F is normal when every irreducible polynomial over F with a root in E splits completely over E
- Chapter lemma-map-into-algebraic-closure (delivered card L48339): F-maps of algebraic extensions extend into a chosen algebraic closure
- Chapter lemma-algebraically-closed (delivered card L48337): over an algebraically closed field every polynomial splits into linear factors
- Isomorphism invariance: if two subfields of the algebraic closure are F-isomorphic, an element and its image have the same minimal polynomial over F

Source reference. Original reference 09GR: sources/stacks-fields/pinned-original-source.tex, source lines 1128–1128. Prerequisite pointer invoked at 2132; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09GU: sources/stacks-fields/pinned-original-source.tex, source lines 1240–1240. Prerequisite pointer invoked at 2122; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let E/F be an algebraic extension of fields. Let $\overline{F}$ be an algebraic closure of F. The following are equivalent*

(1) *E is normal over F, and*
(2) *for every pair $\sigma, \sigma' \in \mathrm{Mor}_F(E, \overline{F})$ we have $\sigma(E) = \sigma'(E)$.*

Proof. Let $\mathcal{P}$ be the set of all minimal polynomials over F of all elements of E. Set

$$T = \{\beta \in \overline{F} \mid P(\beta) = 0 \text{ for some } P \in \mathcal{P}\}$$

It is clear that if E is normal over F, then $\sigma(E) = T$ for all $\sigma \in \mathrm{Mor}_F(E, \overline{F})$. Thus we see that (1) implies (2).

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Conversely, assume (2). Pick $\beta \in T$. We can find a corresponding $\alpha \in E$ whose minimal polynomial $P \in \mathcal{P}$ annihilates β. Because $F(\alpha) = F[x]/(P)$ we can find an element $\sigma_0 \in \mathrm{Mor}_F(F(\alpha), \overline{F})$ mapping α to β. By Lemma 09GU we can extend σ_0 to a $\sigma \in \mathrm{Mor}_F(E, \overline{F})$. Whence we see that β is in the common image of all embeddings $\sigma : E \to \overline{F}$. It follows that $\sigma(E) = T$ for any σ. Fix a σ. Now let $P \in \mathcal{P}$. Then we can write

$$P = (x - \beta_1) \ldots (x - \beta_n)$$

for some n and $\beta_i \in \overline{F}$ by Lemma 09GR. Observe that $\beta_i \in T$. Thus $\beta_i = \sigma(\alpha_i)$ for some $\alpha_i \in E$. Thus $P = (x - \alpha_1) \ldots (x - \alpha_n)$ splits completely over E. This finishes the proof. $\square$